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Isotopes isolation and Uranium Enrichment methods

Research · June 2015

DOI: 10.13140/RG.2.1.4977.1685

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Notes for Isotopes separation : PART 1

1. **Definition** : Isotope separation is the process of concentrating specific isotopes of a chemical element by removing other isotopes.
2. **Types of isotope separation techniques:**
 - Those based directly on the atomic weight of the isotope.
 - Those based on the small differences in chemical reaction rates produced by different atomic weights.
 - Those based on properties not directly connected to atomic weight, such as nuclear resonances.
3. **Cascading** : All large-scale isotope separation schemes employ a number of similar stages which produce successively higher concentrations of the desired isotope. Each stage enriches the product of the previous step further before being sent to the next stage. Similarly, the tailings from each stage are returned to the previous stage for further processing.
4. **The SWU (separative work unit):** is a complex unit which is a function of the amount of uranium processed and the degree to which it is enriched.

For a three-stream separating element, like the centrifuge, we have feed, product, and waste flows. The magnitude of the flows (in moles per unit time) are designated F, P, and W; the mole fraction of the desired isotope in any stream is designated N,

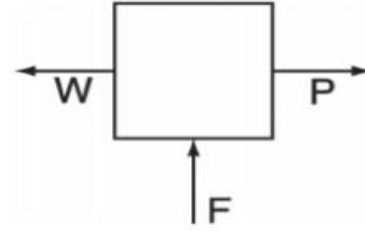


Fig (1.1) Separation process diagram

$$N \equiv \frac{n_A}{n} = \frac{p_A}{p} = \frac{\rho_A}{\rho} \quad (1.1)$$

The separative work unit is given by the following equation:

$$\delta U = PV(N_P) + WV(N_W) - FV(N_F) \quad (1.2)$$

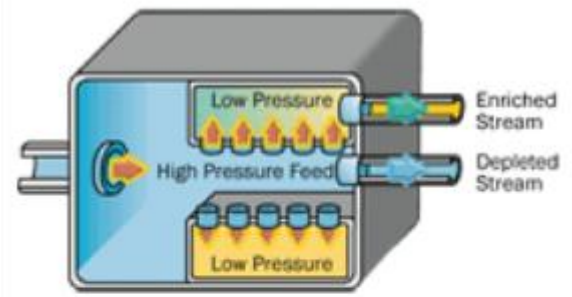
where V is a value function for a binary mixture contrived to operate such that (1) the lowest value is (arbitrarily) set to be a 50:50 mixture, (2) the value increases as the mixture becomes purified in either component, and (3) scales in such a way that the work of n elements (arranged in a non-mixing cascade) will produce n times the work of a single element given the product and waste values $RP = \alpha nRF$ and $RW = \beta nRF$. It can be shown that the value function is:

$$V(N) \equiv (2N - 1) \ln \left(\frac{N}{1 - N} \right) \quad (1.3)$$

5. Enrichment commonly known methods:

I. Gaseous Diffusion

Process: In the gaseous diffusion enrichment plant, uranium hexafluoride (UF_6) gas is slowly fed into the plant's pipelines where it is pumped through special filters called barriers or porous membranes. The holes in the barriers are so small that there is barely enough room for the UF_6 gas molecules to pass through. The isotope enrichment occurs when the lighter UF_6 gas molecules (with the U_{234} and U_{235} atoms) tend to diffuse faster through the barriers than the heavier UF_6 gas molecules containing U_{238} . One barrier isn't enough, though. It takes many hundreds of barriers, one after the other, before the UF_6 gas contains enough U_{235} to be used in reactors. At the end of the process, the enriched UF_6 gas is withdrawn from the pipelines and condensed back into a liquid that is poured into containers. The UF_6 is then allowed to cool and solidify before it is transported to fuel fabrication facilities where it is turned into fuel assemblies for nuclear power plants. The diagram to the right illustrates this gaseous diffusion enrichment process.



The gaseous diffusion process uses molecular diffusion to separate a gas from a two-gas mixture. The isotopic separation is accomplished by diffusing uranium [which has been combined with fluorine to form uranium hexafluoride (UF_6) gas] through a porous membrane (barrier), and using the different molecular velocities of the two isotopes to achieve separation.

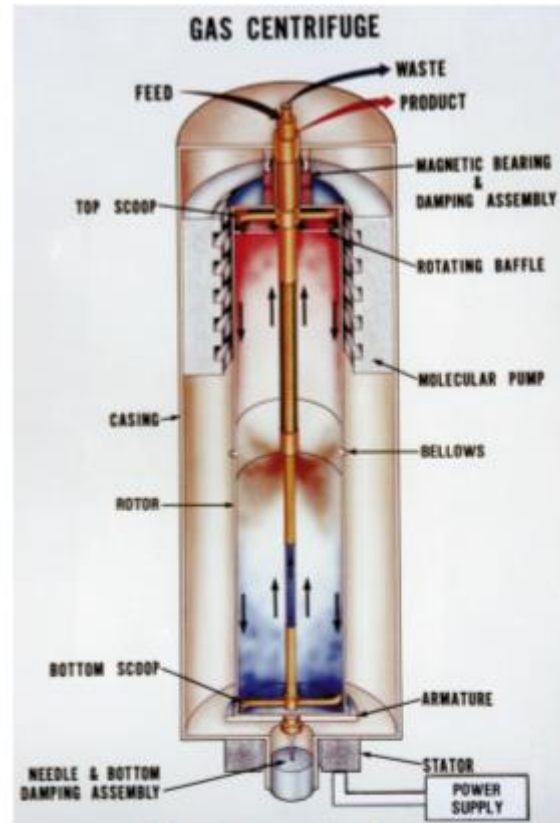
Fig (1.2) Gaseous diffusion

Hazards: The primary hazard in gaseous diffusion plants include the chemical and radiological hazard of a UF_6 release and the potential for mishandling the enriched uranium, which could create a criticality accident (inadvertent nuclear chain reaction).

Plants: The only gaseous diffusion plant in operation in the United States is in Paducah, Kentucky. A similar plant is located near in Piketon, Ohio, but it was shut down in March 2001. Both plants are leased to the United States Enrichment Corporation (USEC) from the U.S. Department of Energy and have been regulated by the NRC since March 4, 1997.

II. Gas Centrifuge

The Centrifuge separation of isotopes was first suggested by Lindemann and Aston (1919), the first experiment at USA was held in 1934 by Prof. Jesse Beams with isotopes of Chlorine, gas centrifuge uranium enrichment process uses a large number of rotating cylinders in series and parallel formations. Centrifuge machines are interconnected to form trains and cascades. In this process, UF_6 gas is placed in a cylinder and rotated at a high speed. This rotation creates a strong centrifugal force so that the heavier gas molecules (containing U_{238}) move toward the outside of the cylinder and the lighter gas molecules (containing U_{235}) collect closer to the center. The stream that is slightly enriched in U_{235} is withdrawn and fed into the next higher stage, while the slightly depleted stream is recycled back into the next lower stage. Significantly more U_{235} enrichment can be obtained from a single-unit gas centrifuge than from a single-unit gaseous diffusion stage. The diagram to the right illustrates this gas centrifuge enrichment process.



Schematic representation of a gas centrifuge; the feed, enriched, and depleted lines are stationary. Centrifugal force causes the heavier ^{238}U molecules to move preferentially to the perimeter of the rotor, which produces partial separation of molecules containing ^{235}U and ^{238}U . This effect is increased by axial countercurrent flow of the gas within the centrifuge.

Fig (1.3) Gas Centrifuge

III. Laser Separation

Isotopic separation of uranium can be achieved based on photo-excitation principles (exciting the molecules using laser light). Such technologies have been named Atomic Vapor Laser Isotope Separation (AVLIS), Molecular Laser Isotope Separation (MLIS), and Separation of Isotopes by Laser Excitation (SILEX). In general, the enrichment process entails using three major systems, which are the laser systems, optical systems, and separation module system. Tunable lasers can be developed to deliver a highly monochromatic radiation (light of a single-color). The radiation from these lasers can photo-ionize a specific isotopic species while not affecting other isotopic species. The affected species is then physically or chemically changed, which enables the material to be separated. AVLIS used a uranium-iron (U-Fe) metal alloy as its feed material, while SILEX and MLIS use UF_6 as its feed material.

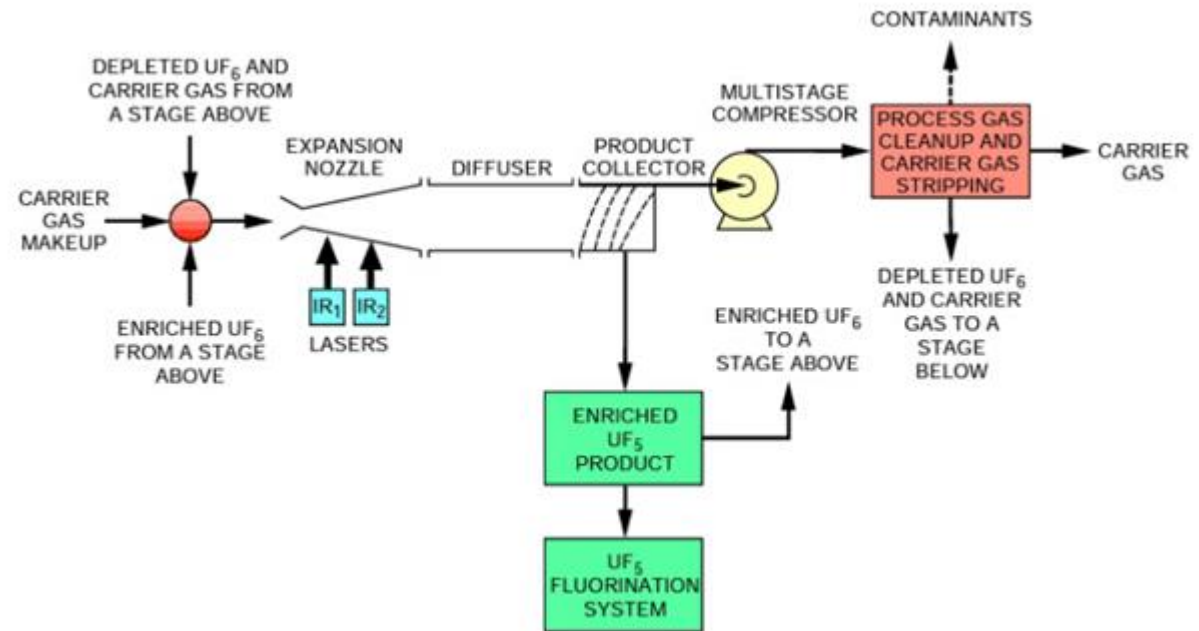


Fig (1.4) Schematic for Laser MILS process

OTHER METHODS:

IV. Aerodynamic processes

Aerodynamic enrichment processes include the Becker jet nozzle techniques developed by E. W. Becker and associates using the LIGA process and the vortex tube separation process. These aerodynamic separation processes depend upon diffusion driven by pressure gradients, as does the gas centrifuge. They in general have the disadvantage of requiring complex systems of cascading of individual separating elements to minimize energy consumption. In effect, aerodynamic processes can be considered as non-rotating centrifuges. Enhancement of the centrifugal forces is achieved by dilution of UF₆ with hydrogen or helium as a carrier gas achieving a much higher flow velocity for the gas than could be obtained using pure uranium hexafluoride. The Uranium Enrichment Corporation of South Africa (UCOR) developed and deployed the continuous Helikon vortex separation cascade for high production rate low enrichment and the substantially different semi-batch Pelsakon low production rate high enrichment cascade both using a particular vortex tube separator design, and both embodied in industrial plant. A demonstration plant was built in Brazil by NUCLEI, a consortium led by Industrial Nucleares do Brasil that used the separation nozzle process. However all methods have high energy consumption and substantial requirements for removal of waste heat; none are currently still in use.

V. Electromagnetic isotope separation

In the electromagnetic isotope separation process (EMIS), metallic uranium is first vaporized, and then ionized to positively charged ions. The cautions are then accelerated and subsequently deflected by magnetic fields onto their respective collection targets. A production-

scale mass spectrometer named the Calutron was developed during World War II that provided some of the U_{235} used for the Little Boy nuclear bomb, which was dropped over Hiroshima in 1945. Properly the term 'Calutron' applies to a multistage device arranged in a large oval around a powerful electromagnet. Electromagnetic isotope separation has been largely abandoned in favor of more effective methods.

VI. Chemical methods

One chemical process has been demonstrated to pilot plant stage but not used. The French CHEMEX process exploited a very slight difference in the two isotopes' propensity to change valency in oxidation/reduction, utilizing immiscible aqueous and organic phases. An ion-exchange process was developed by the Asahi Chemical Company in Japan which applies similar chemistry but effects separation on a proprietary resin ion-exchange column.

VII. Plasma separation

Plasma separation process (PSP) describes a technique that makes use of superconducting magnets and plasma physics. In this process, the principle of ion cyclotron resonance is used to selectively energize the U_{235} isotope in a plasma containing a mix of ions. The French developed their own version of PSP, which they called RCI. Funding for RCI was drastically reduced in 1986, and the program was suspended around 1990, although RCI is still used for stable isotope separation.

6. Cost issues

In addition to the separative work units provided by an enrichment facility, the other important parameter to be considered is the mass of natural uranium (NU) that is needed to yield a desired mass of enriched uranium. As with the number of SWUs, the amount of feed material required will also depend on the level of enrichment desired and upon the amount of U_{235} that ends up in the depleted uranium. However, unlike the number of SWUs required during enrichment which increases with decreasing levels of U_{235} in the depleted stream, the amount of NU needed will decrease with decreasing levels of U_{235} that end up in the DU.

For example, in the enrichment of LEU for use in a light water reactor it is typical for the enriched stream to contain 3.6% U_{235} (as compared to 0.7% in NU) while the depleted stream contains 0.2% to 0.3% U_{235} . In order to produce one kilogram of this LEU it would require approximately 8 kilograms of NU and 4.5 SWU if the DU stream was allowed to have 0.3% U_{235} . On the other hand, if the depleted stream had only 0.2% U_{235} , then it would require just 6.7 kilograms of NU, but nearly 5.7 SWU of enrichment. Because the amount of NU required and the number of SWUs required during enrichment change in opposite directions, if NU is cheap and enrichment services are more expensive, then the operators will typically choose to allow more U_{235} to be left in the DU stream whereas if NU is more expensive and enrichment is less so, then they would choose the opposite.